On Latent Predictive Residuals and Automated Threshold Calibration for Multivariate Time-Series Anomaly Detection: An Empirical Investigation

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Abstract*—Unsupervised anomaly detection in multivariate telemetry requires label-free operational thresholds, typically set via Extreme Value Theory (EVT/SPOT). Reconstructive baselines such as TimesNet and TranAD achieve competitive ranking (PR-AUC 0.29–0.34), but heavy-tailed observation residuals flood operators with false alarms (nominal FPR 24–30%). We investigate whether forecasting in latent representation space via Joint-Embedding Predictive Architectures (JEPAs) conditions residuals for automated thresholding. Across 33 telemetry streams from seven physical domains under held-out calibration (five seeds on the 19-stream subset supplying reported means), unconstrained latent prediction (TS-JEPA) cuts nominal false-positive rates roughly four-fold to 7.6% (paired difference -0.153 vs. TimesNet over all 33 streams, $p = 0.002$). Point-F1 (0.234 vs. 0.197, $p = 0.54$) and PR-AUC (0.323 vs. 0.329, $p = 0.45$) remain statistically indistinguishable, while detection latency increases ($\Delta t = 80$ vs. 48 steps) and strictly causal event recall falls from 58.8% to 46.6%. Three physics-inspired regularizers (Koopman spectral entropy, Reynolds stress, conservative potential flow) yield domain-specific trade-offs rather than universal gains: Koopman operators excel on spacecraft telemetry (PR-AUC 0.591 on SMAP), curl-free priors fail on cyclic kinematics, and no regularizer outperforms unconstrained latent prediction in aggregate ($p \geq 0.61$). EVT calibration also systematically misses its design risk ($q = 10^{-3}$) on non-stationary telemetry, performing comparably to empirical percentiles. Latent prediction thus yields a more conservative detector that suppresses false alarms, but does not improve ranking fidelity or resolve tail-calibration drift.

***Index Terms*—Multivariate Time-Series, Anomaly Detection, Joint-Embedding Predictive Architecture (JEPA), Residual Conditioning, Threshold Calibration, Extreme Value Theory, Empirical Benchmark.**

I. Introduction

Autonomous anomaly detection in multi-sensor telemetry monitors operational health across aerospace systems, Earth observation, cyber-physical infrastructure, and wearable digital healthcare [1], [2]. In remote sensing deployments — NASA SMAP, MSL rovers, industrial IoT facilities, wearable biometric arrays — dozens to hundreds of channels concurrently track power buses, thermal gradients, vibrations, and kinematics [3], and faults must be identified before hardware damage occurs.

A. Challenges in Modeling Multi-Sensor Telemetry

Multi-sensor telemetry reflects a physical state trajectory $\dot{x}(t) = \mathbf{f}(x(t))$, yet modeling dynamics in raw observation space faces four obstacles: estimating derivatives $\dot{x} \approx \Delta x/\Delta t$ on noisy discrete streams amplifies high-frequency noise; telemetry captures only a noisy projection of unmeasured states, violating full-state observability; in large arrays ($K \geq 25$) cross-channel nonlinear interactions resist rigid functional forms; and discrete operational mode transitions (orbital day/night cycling, thruster firings, gait freezing) induce continuous distribution drift.

B. Limitations of Deep Anomaly Detectors

Deep detectors face three limitations. First, generative models (autoencoders, VAEs, masked Transformers) minimize pointwise L_1/L_2 errors in raw signal space [3]–[5], so reconstruction loss allocates capacity to high-frequency jitter rather than nominal dynamics. Second, contrastive frameworks such as NCAD [6] train on synthetic perturbations whose geometry distorts the embedding space, inducing artificial decision boundaries and false alarms. Third, reconstruction errors $\|x - \hat{x}\|_2^2$ accumulate residual variance across dozens of irrelevant channels; under streaming thresholding (SPOT), the resulting heavy-tailed residuals defeat extreme-value calibrators and trigger false-alarm flooding.

C. Investigating Latent Prediction and Inductive Priors

We study whether latent-space prediction mitigates observation-space noise drowning without synthetic contrastive perturbations. A JEPA maps context x_{ctx} into a representation space and predicts target representations z_{tgt} directly [7], [8], discarding unpredictable stochastic jitter and potentially conditioning residuals for tail calibration. Alongside an unconstrained predictor (`TS-JEPA`), we formalize and test three physics-inspired regularizers: *Operator-Entropy JEPA* (spectral-entropy regularization of a latent Koopman transition matrix), *Reynolds-Stress JEPA* (second-moment fluctuation-covariance alignment), and *Potential-Flow JEPA* (curl-free gradient dynamics $\hat{v} = -\nabla\Phi$).

All share a coordinate-saliency scoring gate and a dyadic frequency front-end under SPOT calibration; a strictly causal input-channel gating variant is evaluated separately in a fault-type study.

D. Contributions

This work makes four principal contributions:

1) Quantifying the Latent Residual Conditioning Effect: Forecasting in abstract representation space rather than reconstructing raw waveforms produces lighter-tailed residuals, lowering held-out nominal FPR from 30.0–30.7% to 7.6% (paired -0.153 vs. TimesNet, p=0.002; -0.151 vs. TranAD, p=0.003). Point-F1/PR-AUC remain statistically indistinguishable, detection latency rises (Δt=80 vs. 48 steps), and strictly causal event recall falls (46.6% vs. 58.8% on the multi-seed subset): latent prediction is a conservative low-false-alarm operating point, not universal dominance.

2) Rigorous Evaluation of Physics-Inspired Inductive Priors: Across 33 streams, neither Operator-Entropy ($+0.002$ Point-F1, p=0.71) nor Reynolds-Stress (-0.015, p=0.61) beats unconstrained TS-JEPA, and Potential-Flow is significantly worse (-0.081, p=0.033): Koopman operators excel on quasi-linear spacecraft telemetry (SMAP PR-AUC 0.591) but curl-free flows fail on cyclic kinematics. Fixed-backbone ablations show no significant change when the physical terms are disabled ($p \geq 0.375$).

3) Empirical Analysis of Threshold Calibration Failures: EVT/SPOT systematically misses its design risk (q=10^{-3}) on non-stationary telemetry — nominal FPR 5–8% for JEPAs, $\approx$ 22% for baselines — and kurtosis adjustments degrade rather than correct out-of-sample risk control, performing comparably to simple empirical quantiles.

4) Reproducible Multi-Seed Benchmark: 33 matched streams across seven datasets under a strict held-out calibration protocol (65/17.5/17.5 chronological split with a 64-step isolation gap, up to five seeds, bootstrap CIs, Wilcoxon tests); all code, checkpoints, and logs released.

Section II reviews related work; Section III the benchmark suite; Section IV the formulations; Sections V–VII the protocol, ablations, and results; Section VIII concludes. Supplementary material contains proofs, extended diagnostics, and case-study detail.

II. Related Work

A. Dynamical Systems, Reconstruction, and Forecasting Detectors

Physical modeling of multi-sensor telemetry frames system behavior as continuous trajectories $\dot{x}(t) = \mathbf{f}(x(t))$ [9]. Neural ODEs [10] parameterize velocity fields in raw observation space, where integrating through high-frequency sensor noise induces numerical stiffness and jitter sensitivity. Koopman theory and DMD [11]–[13] instead learn linear operators on observable embeddings; physical disruptions then manifest as spectral decoherence. Our Operator-Entropy variant transfers this idea to a latent transition operator regularized by spectral entropy during training.

The predominant detection paradigm learns nominal patterns via raw-signal reconstruction: LSTM prediction with dynamic thresholding for NASA telemetry [3], association-discrepancy attention in Anomaly Transformer [4], 2D FFT-periodicity decomposition in TimesNet [5], and adversarial sub-series attention in TranAD [14]. These share a vulnerability: point-level losses bias networks toward fitting stochastic noise, and sustained anomalies contaminate the context window. These errors accumulate residual variance across irrelevant channels in high-dimensional arrays.

B. Threshold Calibration, Contrastive, and Self-Supervised Detection

Unsupervised detectors require an automated rule converting residual scores to binary alarms without test labels. Nonparametric dynamic thresholding [3] and Streaming Peaks-Over-Threshold (SPOT) [15]—fitting a Generalized Pareto tail above a high quantile under design risk q—are standard; however, EVT's asymptotic guarantees assume i.i.d. tail samples, which telemetry autocorrelation and operational drift violate.

To avoid reconstructing noise, contrastive methods learn discriminative representations: NCAD [6] trains TCNs against injected synthetic anomalies (which distort the physical manifold and miss subtle real faults), and DCdetector [16] maximizes cross-scale consistency via dual-attention contrasting. Joint-Embedding Predictive Architectures [7], [8], [17] instead predict target representations in latent space, with non-contrastive collapse prevention [18]–[20]. Time-series extensions [21] typically evaluate ranking metrics or post-hoc threshold sweeps without studying how latent residuals behave under automated tail calibration—the gap this work addresses.

C. Physics-Inspired Inductive Biases

PINNs [22] and Hamiltonian/Lagrangian networks [23], [24] embed dynamics as hard constraints; their stiff competing gradients [25] and closed-form-PDE requirements fit poorly to uncalibrated multi-sensor telemetry. We instead test whether *inspired* structures—spectral entropy, second-moment closure, conservative potential flow—improve latent residuals.

D. Clarification of Technical Terminology

- **Physics-inspired vs. physics-informed:** the latter enforces known governing equations as loss residuals; the former adopts physics-motivated structures as exploratory regularizers, without claiming latent coordinates are physical state variables.
- **Causal preprocessing vs. causal decision-time access:** the former computes features strictly from history ($\partial \mathbf{g}/\partial \mathbf{x}_{\text{tgt}} \equiv \mathbf{0}$); the latter specifies whether alarms are issued with 0-lookahead or after an S-step buffer.
- **Conservative vs. dissipative:** autonomous gradient flows $\dot{z} = -\nabla\Phi$ satisfy $\frac{d}{d\tau}\Phi \leq 0$ and preclude periodic orbits.
- **Operator non-degeneracy vs. representation anti-collapse:** the transition operator $\mathbf{K}$ not collapsing to zero is distinct from the encoder not collapsing representations to a constant.

E. Positioning

This investigation bridges joint-embedding predictive learning and multivariate anomaly detection: forecasting latent z_{tgt} rather than reconstructing $\hat{x}$ avoids noise drowning; three candidate physics-inspired regularizers are evaluated empirically against unconstrained latent controls; latent coordinate-saliency gating and dyadic frequency shells preprocess residuals; and EVT/SPOT calibration is evaluated on strictly held-out nominal telemetry to quantify the gap between design risk and empirical false-alarm rates.

III. Multi-Sensor IoT and Remote Sensing Benchmark Suite

Our benchmark suite comprises 11 datasets spanning 45 continuous telemetry streams sourced from NASA spacecraft archives [3], the UCI Machine Learning Repository [26], [27], TimeEval [28], and mTSBench [29]: 33 streams are evaluated under the primary multi-seed held-out calibration protocol and 12 reserved for operational case studies. Streams span four physical domains: *spacecraft telemetry* (SMAP Earth-observation `A-1`/`A-2`/`A-7`/`D-3`/`P-3`/`E-1` power, thermal, and attitude subsystems; MSL rover `C-1`/`C-2`/`D-14`/`D-15`/`D-16`/`M-1` actuator and environmental channels, with two short-nominal streams reserved; SWAN-SF solar magnetograms [30]); *wearable kinematics* (six Daphnet FoG streams [26]: 64 Hz tri-axial accelerometry for Freezing-of-Gait events, so S=64 is a 1-s horizon); *industrial processes* (14 GHL gasoil heating-loop streams, the Genesis pick-and-place testbed, and GECCO water-distribution chemistry/hydrology at 1-minute resolution); and *smart infrastructure* (Room-Occupancy environmental sensing [31], six CICIDS edge network-flow streams, CalIt2 people-flow [27], and Metro I-94 highway traffic).

Each stream provides nominal training sequences and labeled anomaly test sequences; channels are standardized by training statistics with $\boldsymbol{\sigma}$ floored at 10^{-2} against flatline phases. We evaluate the standardized subsets studied in the literature [3], [28] (of 55 SMAP and 27 MSL archive entities) for strict matching and reproducible runtime.

To eliminate circular in-sample calibration, each stream's nominal sequence is partitioned chronologically into training (65%), calibration (17.5%), and held-out evaluation (17.5%) blocks separated by an isolation gap (Section V-A). The held-out protocol spans 33 streams across seven datasets (6 Daphnet, 14 GHL, 4 MSL, 6 SMAP, Genesis, Metro, SWAN); streams whose nominal blocks are too short for disjoint partitions (CalIt2, Room-Occupancy, two MSL) or where multi-seed stride-1 evaluation over 10^6 steps was prohibitive (GECCO, CICIDS) are reserved as case studies.

Supplementary Table S13 specifies the suite: 2,079,507 nominal training and 2,986,574 test timesteps (>5.06M observations), with per-dataset domain, channel dimension K, sequence lengths, contamination ratio $\pi\%$, anomaly typologies, and partition protocols.

A machine-readable manifest (`reports/stream_manifest.csv`, generated by `scripts/build_stream_manifest.py`) enumerates every stream's identifier, channel dimension, split lengths, held-out anomaly counts, benchmark role, and evaluated seed set as recorded in the released artifacts. All counts in this section, the abstract, and the results tables reconcile against it: 33 primary held-out streams (19 non-GHL at five seeds, 14 GHL at three to five seeds depending on size) and 12 case-study streams.

IV. Methodology: Latent Representation Forecasting and Structured Inductive Priors

This section formalizes the mathematical formulation, latent predictors, and inductive biases of the proposed framework.

A. Problem Formulation and Latent Space Prediction

Let $\mathbf{X} \in \mathbb{R}^{T\times K}$ denote a multivariate stream partitioned into context $x_{\text{ctx}} = \mathbf{X}[t-C:t,:] \in \mathbb{R}^{C\times K}$ and suspect horizon $x_{\text{tgt}} = \mathbf{X}[t:t+S,:] \in \mathbb{R}^{S\times K}$. Reconstructive baselines (TimesNet, TranAD, Anomaly Transformer) minimize pointwise error $\min_\theta \mathbb{E}\|\mathbf{X} - \mathcal{D}_\theta(\mathcal{E}_\theta(\mathbf{X}))\|_F^2$, where ambient jitter dominates the Frobenius norm and capacity is spent reconstructing noise (noise drowning). We instead adopt the JEPA framework [7]: a context encoder E_θ maps $x_{\text{ctx}} \to z_{\text{ctx}} \in \mathbb{R}^D$ and an EMA momentum target encoder E_ϕ maps $x_{\text{tgt}} \to z_{\text{tgt}}$, with $\phi_k \leftarrow m_k\phi_{k-1} + (1-m_k)\theta_k$, $m_k \in [0.996, 0.9995]$. The objective forecasts target representations in latent space, testing whether structured priors condition prediction residuals for tail calibration.

B. Latent Predictor Architectures and Dynamical Inductive Priors

To examine whether incorporating structured dynamical inductive biases stabilizes representation forecasting, we evaluate an unconstrained latent predictor alongside three candidate formulations inspired by physical dynamical principles:

1) Operator-Entropy JEPA (Spectral Dynamics & Modal Dispersion): Motivated by Koopman operator theory [32], [33], in which nonlinear trajectories are analyzed via linear operators on observable functions, we parameterize the predictive latent transition via a learned operator $\mathbf{K} \in \mathbb{R}^{D\times D}$: $\hat{z}_{\text{tgt}} = \mathbf{K}z_{\text{ctx}}$. To prevent collapse onto a low-rank subspace (dimensional collapse) or to zero (energy collapse), the regularizer decouples spectral uniformity from total operator energy.

Let $\sigma_1 \geq \cdots \geq \sigma_D \geq 0$ denote the singular values of $\mathbf{K}$, $\lambda_i = \sigma_i^2$ the eigenvalues of $\mathbf{K}\mathbf{K}^T$, and $\mathcal{E}(\mathbf{K}) = \|\mathbf{K}\|_F^2 = \sum_i \lambda_i$ the total energy. For $\mathcal{E}(\mathbf{K}) > 0$, the scale-free spectral distribution over modes is $p_i = \sigma_i^2/\|\mathbf{K}\|_F^2$, $\sum_i p_i = 1$. We omit isotropic diagonal loading from the Gram matrix before normalization — it would give $\mathbf{K} \to \mathbf{0}$ an artificial identity spectrum with maximal entropy, spuriously rewarding collapse. The Von Neumann spectral entropy uses the convention $0\log 0 = 0$:

$$\mathcal{H}_{\text{spectral}}(\mathbf{K}) = -\sum_{i:p_i>0} p_i \log p_i. \tag{1}$$

The *Spectral Entropy Deficiency* quantifies the Kullback-Leibler divergence from the uniform modal equipartition $\mathbf{u} = [1/D, \ldots, 1/D]^T$:

$$\Delta\mathcal{H}(\mathbf{K}) = \log D - \mathcal{H}_{\text{spectral}}(\mathbf{K}) = \mathcal{D}_{\text{KL}}(\mathbf{p} \,\|\, \mathbf{u}) \geq 0. \tag{2}$$

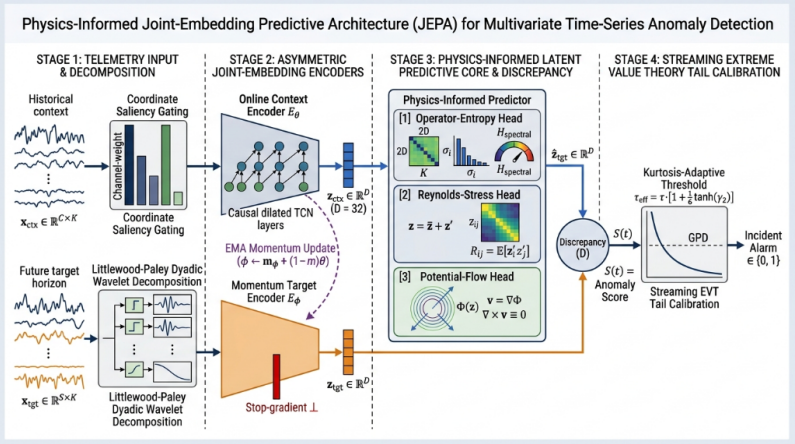


Fig. 1. JEPA telemetry-monitoring architecture: context and suspect windows are encoded into z_{ctx}, z_{tgt} and compared in latent space (avoiding observation-space false-alarm flooding), under three candidate physical regularizers (Koopman spectral entropy, covariance closure, potential flow) with held-out SPOT calibration.

Here, $\Delta\mathcal{H} = 0$ if and only if all modes carry equal energy ($\sigma_1 = \cdots = \sigma_D$), while dimensional collapse to rank 1 yields maximal deficiency $\Delta\mathcal{H} = \log D$.

To penalize energy collapse ($\|\mathbf{K}\|_F \to 0$) and prevent explosive gain, we introduce a non-negative Bregman energy barrier relative to nominal isometric scale $\mathcal{E}_0 = \gamma D$ ($\gamma = 1.0$):

$$\Phi_{\text{energy}}(\mathbf{K}) = r_{\text{safe}}(\mathbf{K}) - 1 - \log r_{\text{safe}}(\mathbf{K}) \geq 0, \quad r_{\text{safe}}(\mathbf{K}) = \max\left(\frac{\|\mathbf{K}\|_F^2}{\gamma D}, \epsilon\right), \tag{3}$$

where $\epsilon = 10^{-6}$ provides a numerical floor. For the identity operator $\mathbf{K} = \mathbf{I}$, $\|\mathbf{I}\|_F^2 = D$, yielding $\Phi_{\text{energy}}(\mathbf{I}) = 0$ exactly. We emphasize that the implemented barrier is *floored*: it attains a finite maximum penalty $\epsilon - 1 - \log\epsilon \approx 12.8$ (before β_{norm} weighting) throughout $\|\mathbf{K}\|_F^2 < \epsilon\gamma D$, rather than diverging; divergence to $+\infty$ is a property only of the unfloored idealization $\Phi^*_{\text{energy}} = r - 1 - \log r$. The operator is initialized near identity ($\mathbf{K} \sim \mathbf{I} + \mathcal{N}(\mathbf{0}, 0.01^2\mathbf{I})$), so $r_{\text{safe}} \approx 1$ and training starts with a vanishing energy penalty; since LayerNorm gives $\mathbb{E}[\|z\|_2^2] \approx D$, the isometric scale $\mathcal{E}_0 = \gamma D$ matches the natural energy of unitary transitions. In sensitivity checks across $\lambda_{\text{ent}} \in \{0.1, 0.5, 1.0\}$ the model converged stably, with 0.5 balancing spectral spread against predictive regression. The composite spectral regularizer is:

$$\mathcal{R}_{\text{spectral}}(\mathbf{K}) = \Delta\mathcal{H}(\mathbf{K}) + \beta_{\text{norm}}\Phi_{\text{energy}}(\mathbf{K}), \tag{4}$$

with $\beta_{\text{norm}} = 1.0$. The total training objective is:

$$\mathcal{L}_{\text{Op-Entropy}} = \|\mathbf{K}z_{\text{ctx}} - z_{\text{tgt}}\|_2^2 + \lambda_{\text{ent}}\mathcal{R}_{\text{spectral}}(\mathbf{K}), \tag{5}$$

with $\lambda_{\text{ent}} = 0.5$ in all experiments. All shared representation components (saliency gating, dyadic frequency shells, LayerNorm) are held fixed across models and controls, isolating the transition-operator loss. We emphasize that $\mathcal{R}_{\text{spectral}}(\mathbf{K})$ acts on the operator matrix, enforcing a non-collapsed operator spectrum; it provides no analytical guarantee against encoder representation collapse, which is mitigated empirically by the EMA target encoder and stop-gradient operations [8], [34].

Proposition 1 (Operator Triviality Invariance)**.** *Let* $\mathcal{L}(\mathbf{K}) = \frac{1}{B}\sum_{b=1}^{B}\|\mathbf{K}z_{ctx}^{(b)} - z_{tgt}^{(b)}\|_2^2 + \lambda_{ent}\mathcal{R}_{spectral}(\mathbf{K})$ *with* $\lambda_{ent} > 0, \beta_{norm} > 0$*, non-trivial target batch* $\frac{1}{B}\sum_{b=1}^{B}\|z_{tgt}^{(b)}\|_2^2 = E_{tgt} > 0$*, and non-degenerate context–target coupling* $\frac{1}{B}\sum_{b=1}^{B} z_{tgt}^{(b)} z_{ctx}^{(b)^T} \neq \mathbf{0}$*. Then (i) the zero operator* $\mathbf{K} = \mathbf{0}$ *is strictly not a local or global minimizer of* $\mathcal{L}(\mathbf{K})$ *under the implemented floored barrier, and (ii) under the unfloored idealization* ($\epsilon \to 0^+$, $\Phi^*_{energy} = r - 1 - \log r$)*, additionally* $\lim_{\|\mathbf{K}\|_F \to 0}\mathcal{L}(\mathbf{K}) = +\infty$.

(Proof in Supplementary Section S-1.)

Remark (Operator Non-Degeneracy vs. Representation Collapse). Proposition 1 concerns only the transition operator: it does not prevent encoder collapse. If E_θ mapped all inputs to a constant $z^{(b)} = \mathbf{c}$, then $\mathbf{K} = \mathbf{I}$ would attain zero prediction error and zero spectral penalty while encoding no information; prevention of that failure mode rests empirically on the stop-gradient momentum dynamics, not on Proposition 1.

2) Reynolds-Stress JEPA (Second-Moment Closure Head): Reynolds decomposition splits a turbulent velocity field into a mean flow and a fluctuation component, where the Reynolds stress tensor captures the second moment of velocity fluctuations. We adapt this second-moment formulation as a latent-space inductive bias; the latent coordinates are learned embeddings, and $\boldsymbol{\Sigma}$ is not a physical momentum-transport tensor.

A two-layer MLP predicts the conditional mean target latent $\hat{z}_{\text{tgt}}$ from z_{ctx}, and a learned linear projection $\mathbf{P} \in \mathbb{R}^{d_s \times D}$ ($d_s = 16$) applied to the EMA target representation yields the empirical second-moment matrix:

$$\boldsymbol{\Sigma}_{\text{obs}} = (\mathbf{P}z_{\text{tgt}})(\mathbf{P}z_{\text{tgt}})^T + 10^{-3}\mathbf{I} \in \mathbb{S}_{++}^{d_s \times d_s}. \tag{6}$$

Because $\boldsymbol{\Sigma}_{\text{obs}}$ is computed from an instantaneous latent vector rather than averaged over the target horizon S, it forms a rank-one outer product with a ridge term enforcing membership in $\mathbb{S}_{++}^{d_s \times d_s}$; statistically it is an auxiliary second-moment projection penalty. The closure head $h_\psi(z_{\text{ctx}})$ outputs a lower-triangular $\mathbf{L} \in \mathbb{R}^{d_s \times d_s}$ with positive diagonal $L_{ii} = \text{softplus}(a_{ii}) + \epsilon$, and the predicted covariance is reconstructed via Cholesky factorization:

$$\boldsymbol{\Sigma}_{\text{pred}} = \mathbf{L}\mathbf{L}^T + 10^{-3}\mathbf{I} \in \mathbb{S}_{++}^{d_s \times d_s}, \tag{7}$$

confining predictions to $\mathbb{S}_{++}^{d_s \times d_s}$. The training objective is:

$$\mathcal{L}_{\text{Reynolds}} = \text{MSE}(\hat{z}_{\text{tgt}}, z_{\text{tgt}}) + 0.1\ \text{MSE}(\boldsymbol{\Sigma}_{\text{pred}}, \boldsymbol{\Sigma}_{\text{obs}}). \tag{8}$$

At inference time, the anomaly score combines the mean latent discrepancy and the fluctuation tensor residual via the

coordinate-saliency gate (formalized in Section IV-C). The stress term enters both the training loss and the test score; its downstream contribution is evaluated in Section VII-B and Table I(a).

3) Potential-Flow JEPA (Conservative Field Theory and Flow-Matching Derivation): Under Helmholtz-Hodge decomposition, a smooth field $v \in C^1(\mathbb{R}^D, \mathbb{R}^D)$ splits into a curl-free potential gradient and a solenoidal field, $v = -\nabla\Phi + \nabla \times A$. As dissipative systems relax toward local minima on potential-energy surfaces, we model latent evolution as the dissipative gradient flow $\dot{z}(\tau) = -\nabla_z \Phi_\psi(z(\tau))$ for a scalar potential network Φ_ψ with smooth (SiLU) activations.

Proposition 2 (Absence of Periodic Attractors in Gradient Flows)**.** *Let $\dot{z} = -\nabla\Phi(z)$ be an autonomous gradient flow on $\mathbb{R}^D$ where $\Phi \in C^2(\mathbb{R}^D, \mathbb{R})$. There exist no non-constant periodic orbits.*

(Proof in Supplementary Section S-1.)

a) Continuous Velocity Field and Multi-Subspace Regime Routing: The velocity field is the negative gradient of a parameterized scalar potential, $\hat{v}(z_t, t, z_{\text{ctx}}, p_{\text{regime}}) = -\nabla_{z_t} \Phi_\psi(z_t, t, z_{\text{ctx}}, p_{\text{regime}})$, where Φ_ψ is a three-layer residual MLP (LayerNorm, SiLU) conditioned on state z_t, diffusion time $t \in [0, 1]$ (sinusoidal embedding), context z_{ctx}, and a discrete regime projection p_{regime}; the raw gradient is tangent-projected via $v_\perp = v - \langle u, v\rangle u$, $u = z_t/\|z_t\|_2$, bounding integration curvature.

Discrete operational modes use $K_r = 4$ learnable orthonormal subspace projections $\mathbf{U}_k \in \mathbb{R}^{D\times d}$ (d=8, D=32); the active regime is selected by maximal captured context energy, $k^* = \arg\max_k \|\mathbf{U}_k^\top z_{\text{ctx}}\|_2^2$, $p_{\text{regime}} = \mathbf{U}_{k^*}\mathbf{U}_{k^*}^\top z_{\text{ctx}}$, with an orthogonality separation loss $\mathcal{L}_{\text{subspace}} = \frac{2}{K_r(K_r-1)} \sum_{i<j} \|\mathbf{U}_i^\top \mathbf{U}_j\|_F^2$. Section VI-A includes a single-regime control (K_r=1) with routing disabled.

b) Flow-Matching Objective, Numerical Midpoint Evaluation, and Dynamical Misspecification: Proposition 2 governs the autonomous continuous flow $\dot{z} = -\nabla\Phi(z)$; the empirical objective instead trains $\hat{v}$ via conditional flow matching (OT-CFM) along linear paths between noise $z_{\text{noise}} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and z_{tgt}: $z_t = (1-t)z_{\text{noise}} + t z_{\text{tgt}}$, $t \sim \mathcal{U}(0, 1)$, with target field $v_t^* = z_{\text{tgt}} - z_{\text{noise}}$ and loss

$$\mathcal{L}_{\text{Pot-Flow}} = \mathbb{E}_{t, z_{\text{noise}}} \left[\|\hat{v}(z_t, t, z_{\text{ctx}}, p_{\text{regime}}) - (z_{\text{tgt}} - z_{\text{noise}})\|_2^2 \right]. \tag{9}$$

Along the deterministic mean trajectory ($z_{\text{noise}} = \mathbf{0}$), $z_t = t\, z_{\text{tgt}}$ and $v_t^* = z_{\text{tgt}}$ is constant.

At inference, rather than integrating an ODE over $t \in [0, 1]$, the detector queries the field at the trajectory midpoint:

$$\mathbf{e}(t) = \hat{v}(0.5 z_{\text{tgt}}, 0.5, z_{\text{ctx}}, p_{\text{regime}}) - z_{\text{tgt}}. \tag{10}$$

Since the reference velocity along the mean path is $v_{0.5}^* = z_{\text{tgt}}$, this midpoint query directly measures the discrepancy between the learned potential gradient and the nominal target, matching the dimensionality (D=32) and scale of the unconstrained TS-JEPA residual. The anomaly score applies the saliency gate: $\mathcal{S}(t) = g(\mathbf{e}(t))$.

Dynamical Misspecification on Non-Conservative Telemetry: A curl-free velocity field ($\nabla \times \hat{v} \equiv \mathbf{0}$) forces circulation along any closed path $\mathcal{C}$ to vanish, $\oint_{\mathcal{C}} \hat{v} \cdot d\mathbf{r} = 0$. Valid for purely dissipative equilibria, this is dynamically misspecified for the non-conservative cyclic dynamics common in telemetry (motor revolutions, diurnal GECCO consumption cycles, periodic Daphnet gait kinematics): normal circulating orbits are penalized as anomalous. This predicts degraded detection on cyclic telemetry, tested in Section VII-B and Supplementary Section S-4.

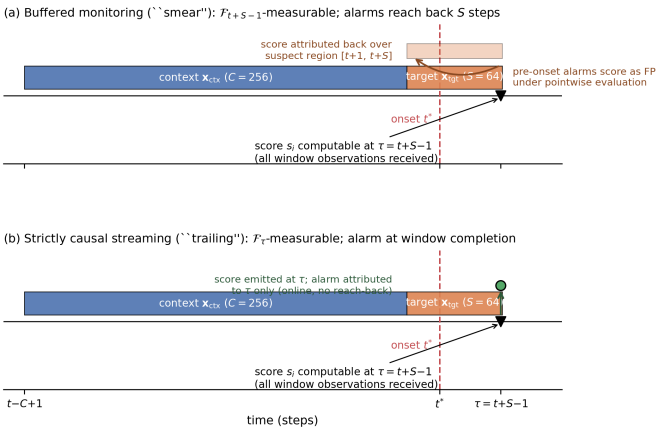


Fig. 2. Score-emission and alarm-attribution timeline. Every score s_i is computable only at window completion $\tau = t+S-1$. (a) Buffered attribution assigns s_i retrospectively to $[t+1, t+S]$; alarms may precede the onset t^* and are penalized as false positives by pointwise scoring. (b) Causal attribution emits s_i at τ only, using no future information at emission time.

C. Anomaly Scoring and Operational Detection Latency

During inference, context windows $x_{\text{ctx}} = \mathbf{X}[t - C : t, :]$ and suspect windows $x_{\text{tgt}} = \mathbf{X}[t : t + S, :]$ are processed at stride 1. For TS-JEPA, Operator-Entropy, and Potential-Flow, the score gates the latent discrepancy through $g(\cdot)$ (Section IV-D1): $\mathcal{S}(t) = g(\hat{z}_{\text{tgt}}(t) - z_{\text{tgt}}(t))$. For Reynolds-Stress, $\mathcal{S}(t) = g(z_{\text{tgt}} - \hat{z}_{\text{tgt}}) + 0.5\, g(\text{vec}(\mathbf{\Sigma}_{\text{obs}} - \mathbf{\Sigma}_{\text{pred}}))$. All shared representation stages are held fixed across variants and ablation controls (Section VI-A).

Operational Distinction: Zero-Lookahead Causal Filtering vs. Buffered Predictive Early Warning. Two regimes exist, formalized in Figure 2: (1) strictly causal streaming (*trailing* attribution), in which the score emitted at τ is $\mathcal{F}_\tau$-measurable — it uses only observations through window completion; and (2) buffered prospective monitoring (*smear* attribution), in which the identical window score is attributed retrospectively over the suspect region $[t+1, t+S]$, so an anomaly at t^* can raise discrepancy up to S=64 steps before onset. Under strict unadjusted point-level evaluation (stride 1), warnings during $[t^* - S, t^*]$ score as false alarms — predictive foresight incurs a retrospective metric penalty for early warning. The two regimes are *not* interchangeable evidence; we report them separately throughout (Tables II, III). Incident analysis (Supplementary Section S-5) also evaluates *Middle* alignment ($t + S/2$); all baselines share identical windowing, horizon, and standardization.

D. Shared Internal Inductive Biases

To stabilize learning across high-dimensional, noisy telemetry, all three architectures incorporate three inductive mechanisms:

1) Coordinate Saliency Gating and the Auxiliary Channel-Gate Variant: On high-dimensional streams ($K \geq 25$), faults frequently originate in a localized subsystem while the remaining channels exhibit ambient noise, diluting localized anomalies in isotropic representations. The deployed detector mitigates this in latent space via an anti-drowning coordinate saliency gate on error vectors $\mathbf{e} = z_{\text{tgt}} - z_{\text{pred}} \in \mathbb{R}^D$:

$$\mathcal{D}_{\text{sal}}(\mathbf{e}) = (1-\alpha)\|\mathbf{e}\|_2 + \alpha\sqrt{D\sum_{d=1}^{D} w_d e_d^2}, \quad w_d = \frac{\exp(|e_d|/\tau)}{\sum_{j=1}^{D}\exp(|e_j|/\tau)}, \tag{11}$$

amplifying sparse coordinate deviations by up to $\sqrt{D}$ while remaining exactly $\|\mathbf{e}\|_2$ under isotropic errors. This is the discrepancy gate used in every experiment: input channels are encoded unweighted, and the gate acts only on the D-dimensional predictive residual.

a) Auxiliary Input-Channel Gate (Evaluated Variant): We also evaluate a strictly causal *Causal Channel Saliency Gate* as an auxiliary variant (full specification, boundary-condition verification, and the fault-type study in Supplementary Section S-3): each channel is weighted by the sigmoid of its standardized context-window variance share, $g_k = \sigma((s_k - \bar{s})/(\sigma_s + \epsilon))$, applied symmetrically to context and target encoders so that $\partial g_k/\partial \mathbf{x}_{\text{tgt}} \equiv \mathbf{0}$ (zero target leakage, with an explicit identity short-circuit at $K{=}1$). Two boundary conditions apply: quiet faults (flatlines, stuck values) cannot be isolated by variance amplification — the fault-type study confirms they are weakly detected under *every* gating regime (recall 0.02–0.07) — and abrupt regime transitions transiently perturb variance shares until the context window refills.

b) Zero-Leakage Data Provenance and Temporal Isolation: We enforce strict temporal isolation across the training, nominal-calibration, and held-out evaluation partitions (Section V-A): standardization statistics come only from the training partition, EVT tail parameters (u, σ, ξ) only from the calibration partition, and evaluation proceeds chronologically with frozen weights and thresholds. The auxiliary gate's weights depend only on x_{ctx}; the deployed pipeline weights inputs uniformly.

2) Multiscale Frequency Front-End: Anomalies span temporal scales from sensor spikes to thermal drifts; the encoder front-end therefore splits each channel into $J{=}4$ smooth dyadic shells on an $N_{\text{fft}} = 256$ rFFT grid, normalized bin-wise into an exact amplitude partition of unity, $\sum_j \Psi_j[\omega_m] \equiv 1$ (measured deviation $< 1.2 \times 10^{-7}$; construction details in Supplementary Section S-4). Each shell feeds a causal 1D convolution ($k \in \{15, 9, 5, 3\}$ coarse-to-fine), recombined by learned scale routing. Because the shells overlap smoothly at band edges, the front-end is *less* frequency-localized than standard wavelet banks: per-band in-octave spectral mass is $\{0.50, 0.33, 0.20, \approx 1.00\}$ versus $\{0.78, 0.50, 0.55, 0.82\}$ (Haar) and $\{0.86, 0.63, 0.64, 0.87\}$ (DB4) (`reports/lp_filterbank_diagnostics.csv`). We therefore describe it as a Fourier-domain multiscale front-end, not an exact Littlewood–Paley octave decomposition, and measure its empirical contribution by matched ablation (Table Ib).

3) Kurtosis-Adaptive Extreme Value Theory (EVT) and Calibration Realities: Binary alarms $\hat{y}_t = \mathbb{I}(\mathcal{S}(t) > \tau_t)$ are generated via Streaming Peaks-Over-Threshold (SPOT) [15]: excesses over a high threshold u (98th percentile) asymptotically follow a Generalized Pareto Distribution (GPD) with scale σ and shape ξ:

$$\tau = u + \frac{\sigma}{\xi}\left[\left(\frac{qN}{N_u}\right)^{-\xi} - 1\right], \tag{12}$$

where $q = 10^{-3}$ denotes the design risk probability, N is total calibration samples, and N_u is the count of excesses. To account for varying tail heaviness across diverse sensor regimes, we modulate the threshold via an empirical heuristic factor governed by the sample excess kurtosis γ_2 of calibration residuals:

$$\tau_{\text{eff}} = \tau \cdot \left[1 + \frac{1}{6}\tanh(\gamma_2)\right]. \tag{13}$$

On heavy-tailed distributions ($\gamma_2 > 0$), τ_{eff} rises to suppress volatile-noise false alarms; on light-tailed ones it tightens for sensitivity.

We emphasize that this is an engineering heuristic for pre-asymptotic finite-sample tails, not an asymptotic EVT theorem. In the controlled calibrator ablation (Section VII-D, Table IV) it does not improve out-of-sample risk control: on held-out data the multiplier raises TS-JEPA nominal FPR from 5.26% to 17.68% with no statistically distinguishable Point-F1 change (0.1694 vs. 0.1577).

More broadly, EVT does not eliminate threshold-calibration difficulty: under $q = 10^{-3}$ (ideal asymptotic FPR 0.1%), measured held-out nominal FPR settles at 5.3–6.2% for the JEPAs and $\approx 22\%$ for TimesNet/TranAD, owing to finite-sample variance, residual autocorrelation, and non-stationary drift between nominal training and deployment. The calibrated finding of this paper is narrower: latent predictive residuals produce substantially fewer false alarms than observation-space residuals under identical calibration, reflecting lighter tails at the cost of lower causal recall. Thresholds use static EVT on held-out nominal splits; dynamic rolling SPOT is evaluated separately in the operational incident analysis.

V. Experimental Protocol and Evaluation Metrics

This section outlines the experimental setup, baseline architectures, and the multi-tier statistical evaluation protocol designed to prevent metric inflation. The master experiment matrix enumerating all evaluation suites (stream counts, train/calibration/test partitions, horizons, calibrators, and metrics) is provided in Supplementary Table S1.

A. Experimental Setup and Hyperparameters

To ensure rigorous and fair comparison, all architectures are trained and evaluated under an identical streaming time-series pipeline. Telemetry streams are processed using a sliding window with past context duration $C = 256$ and suspect target horizon $S = 64$ (total window length $L = 320$), evaluated with a stride of 1 timestep across all testing streams.

Context representations $z_{\text{ctx}} \in \mathbb{R}^D$ ($D = 32$) are extracted using a causal dilated Temporal Convolutional Network (TCN) comprising 3 residual blocks with channel depth $F = 48$, kernel size $K = 5$, and dilation factors $d \in \{1, 2, 4\}$. While the convolutional receptive field across the 3 dilated blocks spans $1 + 2(K-1)\sum_l d_l = 57$ timesteps, global temporal average pooling across the resulting feature maps aggregates information across the entire context window $C = 256$, condensing the sequence into a fixed-dimensional latent representation $z_{\text{ctx}} \in \mathbb{R}^D$ regularized by Layer Normalization and dropout ($p = 0.20$). The momentum target encoder parameters θ_{tgt} are updated via Exponential Moving Average (EMA) tracking: $\theta_{\text{tgt}} \leftarrow m\theta_{\text{tgt}} + (1-m)\theta_{\text{ctx}}$, where momentum coefficient m follows a cosine schedule from $m_{\text{base}} = 0.996$ to $m_{\max} = 0.9995$.

Models are optimized using AdamW with an initial learning rate $\eta = 10^{-3}$ decayed to $\eta_{\min} = 10^{-5}$ via Cosine Annealing, weight decay 10^{-4}, and mini-batch size $B = 32$ across 20 training epochs on nominal telemetry sequences.

a) Held-Out Calibration Protocol: To evaluate out-of-sample threshold calibration without circularity, each stream's nominal telemetry is partitioned into three disjoint chronological blocks: *training* (first 65%; representation weights only), *nominal calibration* (next 17.5%; withheld from training, used solely to fit per-stream GPD tail parameters σ, ξ above u=0.98 at q=10^{-3}, with γ_2 estimated strictly within this block), and *nominal evaluation* (final 17.5%, separated by an unobserved 64-step isolation gap; held-out FPR measured on data the threshold never saw). Thresholds then remain fixed during test evaluation to measure static EVT stability under drift; dynamic rolling SPOT is evaluated separately in the operational incident analysis (Supplementary Section S-5).

b) Calibrator Assignment Across Experiments: To avoid ambiguity, we specify the calibrator per experiment. **Primary held-out benchmark** (Tables II, III, IV; all paired statistics): standard SPOT ($u = p_{98}$, $q = 10^{-3}$, no kurtosis multiplier); headline metrics are the "SPOT" column of Table IV, with calibrator variants compared on identical score streams. **Ablations and controlled studies** (Section VI): SPOT with the $\tanh(\gamma_2)/6$ kurtosis multiplier (the released `adaptive_kurtosis` flag). **Early-warning and EVT mechanism studies**: standard SPOT unless the calibrator is the object of comparison. **Incident metrics** (Supplementary Section S-5): rolling SPOT. No experiment mixes calibrators across models within a table.

c) Multi-Seed Replication, Primary Endpoints, and Multiplicity Controls: Principal models on the 19 non-GHL streams are trained across five seeds (42, 123, 456, 7, 2024); the 14 large GHL streams ($> 10^6$ steps) use three seeds (five on the two smallest) for cost, and ablation controls use three. Two co-primary endpoints: (i) *nominal FPR reduction* on held-out nominal data, testing whether latent prediction stabilizes EVT bounds; and (ii) *detection parity* via unadjusted Point-F1 and PR-AUC, testing whether suppression degrades discrimination.

Inference reports stream-macro means, seed SDs, moving-block bootstrap 95% CIs (block L=64), and two-sided Wilcoxon signed-rank tests. The independent unit of analysis is the telemetry stream (n=19 non-GHL at five seeds; n=33 where GHL enters at available coverage): seeds are averaged within stream and repeated windows aggregated to one stream-level score, so nothing is double-counted. Because nonsignificant tests do not establish equivalence, we make no noninferiority claim: paired differences are reported with effect sizes and CIs for readers to judge practical margins. p-values across comparisons are Benjamini–Hochberg FDR-adjusted at $q_{\text{FDR}} = 0.05$; CIs spanning zero or adjusted $p > 0.05$ are treated as inconclusive, not as equivalence. Supplementary Table S21 maps every reported test to its endpoint, cohort, paired-stream count, seed treatment, raw and adjusted p, and effect-size interval.

B. Baseline Architectures and Parameter Capacities

To isolate algorithmic principles from external implementation discrepancies, all baselines are executed within a unified PyTorch benchmarking pipeline under identical windowing, standardization, and EVT calibration. TimesNet and TranAD are in-house re-implementations of the published architectures (rather than vendor code), configured to published depths and evaluated under the identical optimization budget (AdamW, $\eta = 10^{-3}$, cosine decay, 20 epochs); no method received per-model hyperparameter tuning beyond the shared configuration, and no threshold, early-stopping, or model-selection step accessed held-out anomaly labels. Hardware-specific throughput, batch, warm-up, and per-timestep latency measurements are reported in Supplementary Table S10.

TimesNet [5]: Multi-periodic reconstructive architecture mapping 1D series to 2D variation spaces via FFT period discovery and Inception blocks (648,553 parameters at K=25; 574,529 at K=9).

TranAD [14]: Adversarial reconstructive Transformer with two-stage self-conditioning and sub-series attention (205,746 at K=25; 201,090 at K=9).

TS-JEPA (Unconstrained Control): Standard JEPA predicting z_{tgt} from z_{ctx} via a 2-layer MLP without physical priors (289,061 at K=25; 218,309 at K=9; only the input projection scales with K at +48 parameters per channel).

Physics-Inspired Formulations: Operator-Entropy (285,733 at K=25; its explicit $D \times D$ Koopman matrix replaces the MLP predictor, -1.15% capacity), Reynolds-Stress (295,953, $+2.4\%$ for the covariance closure head), and Potential-Flow (338,117, $+17.0\%$ for the potential network). These specifications confirm performance differences do not stem from parameter scaling.

Linear AR Forecaster (Non-Neural Baseline): Ridge-regularized linear map from flattened context $\mathbf{x}_{\text{ctx}} \in \mathbb{R}^{C \cdot K}$ to target windows $\mathbf{x}_{\text{tgt}} \in \mathbb{R}^{S \cdot K}$, scored by observation-space prediction error under identical held-out EVT — isolating whether latent representation learning is necessary at all.

C. Multi-Tier Statistical Evaluation Metrics

Critical surveys [28], [35] document severe reporting vulnerabilities in this literature — point-adjustment inflation, channel-averaging imbalances, threshold gaming — so we report metrics across complementary tiers:

Dataset-Macro Point-F1 (Primary Metric): Unweighted mean of Point-F1 across datasets:

$$\text{Macro-F1}_{\text{DS}} = \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} \left(\frac{1}{|C_d|} \sum_{c \in C_d} \text{F1}_c \right) \tag{14}$$

where C_d is the channel set of dataset d; this prevents high-stream datasets (GHL, 14 streams) from dominating and weights each physical domain equally.

Pooled Micro Point-F1: Global pooling of TP/FP/FN over all test timesteps, Micro-F1 $= 2\sum_c \text{TP}_c / (2\sum_c \text{TP}_c + \sum_c \text{FP}_c + \sum_c \text{FN}_c)$; it exposes models that flood long streams with false alarms. **Channel-Macro Point-F1** is the standard per-channel average. **PR-AUC** (average precision over all score thresholds) evaluates raw ranking independent of calibration. **Event Recall** is the fraction of contiguous anomaly segments $[t_{\text{start}}, t_{\text{end}}]$ in which at least one alarm fires. **Mean Detection Delay** Δt is the onset-to-first-alarm lag averaged over detected events (undetected events enter via Event Recall, not an arbitrary penalty). **Trivial all-positive reference**: a constant alarm scores $\text{F1}_{\text{all-pos}} = 2\pi/(1+\pi)$ at base rate π; models below it provide less value than an uncalibrated alarm. **Contiguous PA-F1**: point adjustment within contiguous ground-truth segments only; we also report test-sweep PA-F1 (best post-hoc threshold) to quantify the residual calibration gap.

D. Evaluation Regimes: Zero-Lookahead Causal vs. Buffered Inference

To keep early-warning capability separate from future-observation access, we evaluate two regimes: **strictly causal online detection** (at decision time T only $\{X_\tau\}_{\tau \le T}$ is accessible; scores use trailing attribution; zero lookahead) and **buffered window detection** (S=64 future timesteps are buffered before scoring, capturing full trajectory evolution at a deliberate 64-step latency). Both record Mean Detection Delay, Event Recall within delay budgets ($\Delta t \le 10, 64$), and false alarms per 1,000 steps ($\text{FAR}_{1\text{k}}$).

VI. Ablation Studies and Analysis

A. Ablation 1: Physical Regularizer Controls on Fixed Architectures

To test whether the physics-inspired regularizers benefit detection, Table I(a) reports controlled ablations across all 19 non-GHL streams (principal models at five seeds, controls at three, under the held-out protocol), disabling each physical term on a fixed backbone:

1) **Operator-Entropy vs. $\lambda_{\text{ent}} = 0$:** Disabling the spectral regularizer leaves Point-F1 unchanged (0.241 ± 0.014 vs. 0.244 ± 0.024; difference -0.003, 95% interval $[-0.011, +0.005]$, Wilcoxon $p = 0.62$); PR-AUC stays 0.347–0.348.
2) **Reynolds-Stress vs. $\alpha_{\text{stress}} = 0$:** Disabling the stress loss yields 0.203 ± 0.059 vs. 0.208 ± 0.071 ($+0.005$, $[-0.006, +0.016]$, $p = 0.41$); the null is also insensitive to the covariance estimator (temporal/batch covariances shift F1 by ≤ 0.010 over nine streams).
3) **Potential-Flow vs. no-regime (K_r=1):** Disabling regime routing *improves* Point-F1 to 0.129 from 0.096 ($p = 0.048$); both underperform TS-JEPA (0.234, $p = 0.013$).

Across the full 19-stream suite, the physical regularizers produce no statistically detectable improvement over unconstrained latent prediction.

B. Ablation 2: Auxiliary Input-Channel Gating Variant

The auxiliary causal input-channel gate is compared against uniform pooling and non-causal controls under identical TS-JEPA configuration (Supplementary Section S-3, Tables S3, S4). The effect is stream-dependent and macro Point-F1 favors uniform weighting (0.118 vs. 0.077): gating roughly doubles recall on transient faults at $\sim 3\times$ the nominal FPR, while quiet faults remain weakly detected under every regime (recall 0.02–0.07). It is therefore an evaluated variant; the deployed pipeline retains uniform input weighting.

C. Ablation 3: Multiscale Dyadic-Shell Front-End

Replacing the standard front-end with the Fourier-domain dyadic shells plus per-shell causal convolutions and learned routing yields a modest, stream-dependent effect (Table I(b)): macro buffered Point-F1 improves from 0.088 to 0.131 across five multivariate streams, concentrated on MSL and SMD, with a small SMAP regression. Two caveats: the block adds $\approx 1.6 \times 10^5$ parameters, and measured shell overlap means octave isolation is not attained. The front-end is a modest contributor, not the primary driver.

D. Ablation 4: Threshold Calibration Rules on Nominal Splits

The full calibrator comparison is in Section VII-D; two findings bear on the ablations: non-parametric rules match SPOT (validation-max FPR 0.0553 vs. 0.0526 at equal F1), and the kurtosis-adaptive multiplier degrades calibration ($0.0526 \rightarrow 0.1768$ FPR at indistinguishable F1).

E. Controlled Backbone Comparison: Latent Space vs. Observation Space

Does predicting in a contracted latent space condition residuals for EVT thresholding, or is the gain attributable to prediction, compression, or the EMA target encoder? To isolate the mechanism, all 19 non-GHL streams were re-run under the primary held-out protocol (five seeds, identical windows, AdamW budget, and SPOT u=0.98/q=10^{-3} calibration) with five arms sharing the `HybridTCNEncoder` backbone (C=256, S=64, d=32, three dilated TCN layers): **TCN-Obs-Recon** (observation reconstruction), **TCN-Obs-Pred** (observation forecasting), **JEPA-CtxRecon** (latent "prediction" of the EMA-encoded *context* — compression without foresight), **JEPA-NoEMA** (latent prediction with an online hard-copy target), and **TS-JEPA**. Supplementary Table S17 reports the per-arm residual diagnostics (tail quantiles, excess kurtosis, extremal index, ACF, effective exceedance counts); the earlier 15-stream multi-SNR study remains in Supplementary Table S2.

Latent space, not forecasting, conditions the tail: Observation-space objectives produce virtually identical heavy

TABLE I

ABLATION STUDIES. (A) MATCHED-CAPACITY REGULARIZER CONTROLS ON FIXED BACKBONES, 19 NON-GHL STREAMS (POINT-F1 = STREAM-MACRO MEAN ± SEED SD; HELD-OUT FPR ON UNSEEN NOMINAL DATA). (B) DYADIC-SHELL FRONT-END ON/OFF UNDER IDENTICAL TS-JEPA CONFIGURATION. AUXILIARY INPUT-CHANNEL GATING AND FAULT-TYPE RECALL PANELS ARE IN SUPPLEMENTARY SECTION S-3.

(a) Regularizer Controls on Fixed Architectures

Model / Control	Point-F1 (buffered)	PR-AUC	Held-out FPR	Event Recall
TS-JEPA (Unconstrained)	0.234 ± 0.030	0.323	0.076	0.481
Operator-Entropy JEPA	0.237 ± 0.013	0.323	0.086	0.506
$\lambda_{\text{ent}} = 0$ control	0.244 ± 0.024	0.347	0.083	0.518
Reynolds-Stress JEPA	0.207 ± 0.052	0.307	0.078	0.434
$\alpha_{\text{stress}} = 0$ control	0.203 ± 0.059	0.311	0.066	0.414
Potential-Flow JEPA	0.089 ± 0.036	0.201	0.091	0.303
no-regime control ($K_r = 1$)	0.129 ± 0.061	0.265	0.081	0.392
TimesNet (Baseline)	0.197 ± 0.018	0.329	0.300	0.583
TranAD (Baseline)	0.130 ± 0.004	0.287	0.307	0.490

(b) Dyadic-Shell Front-End On/Off

Stream	Standard Conv	Dyadic Shells	Δ
SMAP `P-3`	**0.294**	0.261	−0.033
MSL `M-1`	0.002	**0.052**	+0.050
SMD `machine-1-2`	0.000	**0.177**	+0.177
Daphnet `S01R01E1`	0.144	**0.165**	+0.021
GECCO `water_quality`	0.000	0.001	+0.001
Macro Mean	0.088	**0.131**	+49.6% rel.

tails regardless of direction (TCN-Obs-Recon vs. TCN-Obs-Pred: γ_2 8.31/8.24, held-out FPR 0.303/0.310, extremal index $\theta \approx 0.12$). Latent objectives compress tails ≈8× (γ_2 1.0–1.6, $\theta \approx 0.18$) and cut held-out FPR ≈3×: TS-JEPA -0.230 vs. TCN-Obs-Pred (Wilcoxon p=0.031) and -0.223 vs. TCN-Obs-Recon (p=0.058). Forecasting in raw space does not condition residuals because they directly reflect sensor jitter.

Compression vs. prediction vs. EMA: JEPA-CtxRecon — which never predicts — retains most of the conditioning (-0.203 vs. TCN-Obs-Pred, p=0.055; only -0.028 below TS-JEPA, p=0.23), and removing the EMA target changes little (JEPA-NoEMA: -0.033 vs. TS-JEPA, p=0.17). The threshold-conditioning gain is therefore attributable primarily to *scoring in the contracted latent space*, with the predictive objective and EMA target as secondary contributors. PR-AUC differences across all five arms are inconclusive ($p \geq 0.16$): the effect is residual conditioning, not improved anomaly–nominal separation.

Native Window Horizons: Baselines do not suffer merely from the long W=320 window: at their native horizons tail conditioning *worsens* (Supplementary Table S16). TranAD at W=10 spikes to $\gamma_2 = 448.25$ and TimesNet at W=96 reaches $\gamma_2 = 98.14$ with 44.09% empirical FPR, while TS-JEPA retains $\gamma_2 = 3.14$: raw-space tail corruption is inherent, not a window-length artifact.

VII. RESULTS AND DISCUSSION

A. Benchmark Performance on Continuous Telemetry

Table II reports the comparison across 33 continuous telemetry streams under the held-out calibration protocol of Section V-A (SPOT, $u = 0.98, q = 10^{-3}$). Primary findings:

False Alarm Suppression on Held-Out Telemetry: On the 19 non-GHL streams, latent predictors achieve held-out nominal FPR 0.076–0.086 versus 0.300 (TimesNet) and 0.307 (TranAD). Across all 33 streams the paired reduction is significant: TS-JEPA -0.153 vs. TimesNet (CI $[-0.247, -0.071]$, p=0.002) and -0.151 vs. TranAD (p=0.003); Operator-Entropy shows the same effect (-0.144/-0.142, p=0.006). The mechanism is isolated under the same-backbone control (Section VI-E).

Point-F1 and Continuous Ranking (PR-AUC) Parity: False-alarm suppression does not translate into statistically distinguishable detection gains. On the non-GHL subset TS-JEPA attains 0.234 Point-F1 / 0.323 PR-AUC versus 0.197/0.329 (TimesNet) and 0.130/0.287 (TranAD). Across all 33 streams the paired differences are inconclusive: vs. TimesNet +0.012 F1 (CI $[-0.053, +0.075]$, p=0.54), -0.000 PR-AUC (p=0.39); vs. TranAD +0.051 F1 (p=0.88), +0.023 PR-AUC (p=0.16).

Event Recall and Detection Conservatism: Lower FPR accompanies lower event recall: on non-GHL streams TimesNet captures 58.6% of events and TranAD 48.1%, versus 49.8% (TS-JEPA), 39.8% (Reynolds), 33.1% (Potential-Flow). Subtle onsets that do not shift dominant latent coordinates are filtered with the noise.

Detection Delay Trade-Offs (Δt): Mean detection delay (Table II) varies widely: TimesNet responds fastest (Δt=48 steps), TS-JEPA averages 80, and the physics variants are slower still (133–192); TranAD is slowest (234). Because latent encoders smooth observation noise, alarms trigger only once anomalies propagate into dominant latent coordinates — deployments trade 4× fewer nominal false alarms for added initial latency.

Necessity of Nonlinear Representation Learning (Linear AR Control): A ridge-regularized linear AR forecaster under identical calibration attains FPR 0.006 but collapses to Point-F1 0.057 (precision 4.35%): observation-space linear dynamics cannot separate faults from background variance. Nonlinear latent representation is necessary to combine controlled false alarms with usable detection. Two training-free references sharpen the point (Supplementary Table S20): a PCA-subspace reconstructor matches TimesNet almost exactly (0.311/0.198/0.331 vs. 0.300/0.197/0.329 FPR/F1/PR-AUC), while a per-channel max-z scorer attains the best ranking of any baseline (0.374) at the worst FPR (0.339): ranking quality and threshold conditioning are independent properties.

Domain-Specific Telemetry Dynamics: Relative performance depends on the physical sensing regime (per-dataset breakdown in Supplementary Tables S15–S14): TimesNet leads on Daphnet periodic kinematics (0.245 vs. 0.155 F1), where 2D Fourier decomposition captures gait structure; latent prediction leads on spacecraft telemetry (SMAP: OpEntropy 0.591 PR-AUC; MSL: 0.445 vs. TimesNet 0.159). On GHL and Metro all models struggle (0.000–0.022 F1): subtle actuator drifts and extreme traffic surges evade static EVT bounds.

B. Evaluating Physical Inductive Priors: Hypothesis Testing and Null Results

Does adding physics-inspired regularizers (Koopman spectral entropy, Reynolds-stress closure, curl-free potential flow)

TABLE II
HELD-OUT CALIBRATION BENCHMARK, 33 TELEMETRY STREAMS (7 DOMAINS). THRESHOLDS: SPOT (q=10^{-3}) ON HELD-OUT NOMINAL SPLITS. METRICS: HELD-OUT NOMINAL FPR (ON DATA THE THRESHOLD NEVER SAW), UNADJUSTED POINT-F1, PR-AUC, EVENT RECALL, MEAN DETECTION DELAY Δt. "NON-GHL": STREAM-MACRO OVER 19 STREAMS AT FIVE SEEDS; "GHL": 14 STREAMS AT THREE-TO-FIVE SEEDS; "DATASET-MACRO": UNWEIGHTED MEAN OF THE 7 DATASET MEANS. BEST IN **BOLD**, SECOND UNDERLINED.

	Non-GHL Streams (19 Streams, 5 Seeds)					Summary	Gasoil Heating Loop (14 Streams)	
Model Architecture	**Held-Out FPR ↓**	**Point-F1 ↑**	**PR-AUC ↑**	**Event Recall ↑**	**Mean Delay (Δt) ↓**	**DS-Macro F1 ↑**	**Held-Out FPR ↓**	**Point-F1 ↑**
Latent-Space Predictive Architectures (JEPAs)								
TS-JEPA (Unconstrained Control)	0.076	0.234	0.323	0.481	80.3	**0.148**	0.005	0.000
Operator-Entropy JEPA	0.086	**0.237**	0.323	0.506	192.3	0.130	0.012	0.001
Reynolds-Stress JEPA	0.078	0.207	0.307	0.434	132.8	0.122	**0.003**	0.002
Potential-Flow JEPA	0.091	0.089	0.201	0.303	146.9	0.065	0.011	0.006
Observation-Space Reconstructive Baselines								
TimesNet (ICLR 2023)	0.300	0.197	**0.329**	**0.583**	48.5	0.100	0.061	**0.022**
TranAD (VLDB 2022)	0.307	0.130	0.287	0.490	234.4	0.069	0.046	0.020
Non-Neural Linear Baseline								
Linear AR Forecaster	**0.006**	0.057	0.262	0.478	**0.0**	0.038	**0.002**	0.000

improve detection over unconstrained `TS-JEPA`? The 33-stream data refute this:

1) **Operator-Entropy vs. TS-JEPA:** paired difference $+0.002$ Point-F1 (CI $[-0.025, +0.029]$, p=0.71), $+0.002$ PR-AUC (p=0.86), $+0.009$ FPR (p=0.75); leads on only 9 of 33 streams.
2) **Reynolds-Stress vs. TS-JEPA:** -0.015 Point-F1 ($[-0.042, +0.007]$, p=0.61), -0.007 PR-AUC (p=0.13), $+0.001$ FPR (p=0.92); leads on 11 of 33.
3) **Potential-Flow vs. TS-JEPA:** significantly worse Point-F1 (-0.081, $[-0.160, -0.018]$, p=0.033) and nominally lower PR-AUC (-0.067, p=0.58); leads on 7 of 33.

These findings align with the matched-capacity regularizer controls in Table I(a) and Section VI-A, where disabling each physical loss term on a fixed backbone produces changes well within seed noise. We conclude that the physical regularizers provide no measurable anomaly detection advantage over unconstrained latent prediction. Performance differences between models stem from shared representation conditioning rather than the physical loss terms.

The dynamical diagnostics evaluating *why* each regularizer fails (operator rank preservation without detection gain, covariance-shift insensitivity, and curl-free misspecification on cyclic telemetry) are reported in Supplementary Section S-4 (Tables S5–S7).

C. Operational Incident Evaluation and Causal-vs-Buffered Trade-offs

Event-level incident metrics, latency, and rolling-SPOT case studies are in Supplementary Section S-5: rolling SPOT suppresses false alarms up to $21\times$ on drift-prone streams at a recall cost, and TS-JEPA's causal false-positive rate (4.18%) sits an order of magnitude below reconstructive baselines (20.8–30.1%).

Table III reports the same benchmark under strictly causal (0-lookahead) streaming. On the 19 non-GHL streams TS-JEPA reduces held-out nominal FPR to 0.0136 versus 0.2427 (TimesNet) and 0.2219 (TranAD), an $\approx 18\times$ suppression; on the 14 GHL streams causal FPR falls below 0.009 for JEPAs, but causal Point-F1 there is near-zero for latent predictors ($\leq$ 0.006 vs. 0.022 TimesNet). Causal Point-F1 drops to 0.1118 ± 0.0412 (TS-JEPA) versus 0.2091 ± 0.0265 (TimesNet), and event recall declines from 58.82% to 46.55%: without a lookahead buffer, latent predictive scoring is an ultra-conservative filter that prevents false alarms at the price of detection sensitivity.

a) Buffered Early-Warning Lead Time: Under buffered inference across five streams, TS-JEPA warns pre-onset for 12.3% of events (7 of 57 pooled; median lead 180 steps, 23.8 false warnings per 1,000 nominal steps), while TimesNet warns for 73.7% at a comparable lead (204 steps) but emits 206.4 false warnings per $1{,}000$ — on GECCO approaching a constant-alarm regime (873.6/1k). Pre-onset coverage is confounded by the underlying alarm rate: TS-JEPA trades early-warning coverage for an order-of-magnitude lower nuisance rate.

b) Detection Quality at Matched False-Alarm Budgets: Supplementary Table S18 re-evaluates all models at common held-out FPR budgets $q^* \in \{0.001, 0.005, 0.01, 0.05\}$, removing the confound that native SPOT thresholds land on different effective operating points. Under strictly causal streaming the ordering partially reverses: at q^*=0.05, TS-JEPA attains event recall 0.774 versus 0.618 (TimesNet) while emitting 180 versus 262 false alarms per $1{,}000$ nominal steps — the Table III gap largely reflects TimesNet's miscalibrated threshold admitting a higher realized alarm rate, not an intrinsic sensitivity deficit. Under buffered attribution the ordering remains mixed (0.486–0.560 vs. 0.583–0.612 recall), while TS-JEPA consistently emits fewer nuisance alarms per nominal step; pre-onset conventions are reported per operating point.

D. Automated Threshold Calibration Realities on Non-Stationary Telemetry

Does automated extreme-value thresholding adhere to its design risk q out-of-sample? Table IV compares calibrators on identical score streams across all 33 streams under buffered inference. Four realities emerge:

1) **Systematic design-risk overshoot:** SPOT (u=$0.98, q$=10^{-3}) never achieves its design risk out-of-sample — held-out nominal FPR averages 5.26% (TS-JEPA), 6.17% (OpEntropy), 22.17% (TimesNet), 22.17% (TranAD), a 53–$222\times$ overshoot.
2) **High failure rate:** 95% bootstrap CIs exclude q in $61\%/65\%/86\%/82\%$ of runs (TS-JEPA/OpEntropy/TimesNet/TranAD); non-stationary drift violates EVT's stationary-tail assumptions.

TABLE III
STRICTLY CAUSAL STREAMING BENCHMARK (0-LOOKAHEAD, TRAILING ATTRIBUTION), 33 STREAMS, HELD-OUT SPOT (q=10^{-3}). "NON-GHL": 19 STREAMS AT FIVE SEEDS (STREAM-MACRO, $\pm$ SEED SD); "GHL": 14 STREAMS AT THREE-TO-FIVE SEEDS. LATENT PREDICTORS CUT NOMINAL FALSE ALARMS TO 0.97%–1.89% ON NON-GHL (12–25$\times$ BELOW RECONSTRUCTIVE BASELINES) AT THE COST OF LOWER CAUSAL POINT-F1 AND EVENT RECALL.

	Non-GHL Streams (19 Streams, 5 Seeds)					Gasoil Heating Loop (14 Streams)		
Model Architecture	**Held-Out FPR ↓**	**Point-F1 (Mean ± SD) ↑**	**PR-AUC ↑**	**Event Recall ↑**	**Test FPR ↓**	**Held-Out FPR ↓**	**Point-F1 ↑**	**Event Recall ↑**
Latent-Space Predictive Architectures (JEPAs)								
TS-JEPA (Unconstrained Control)	0.0136	0.1118 ± 0.0412	0.2672	0.4655	0.0844	0.0035	0.000	0.000
Operator-Entropy JEPA	0.0189	0.1093 ± 0.0085	0.2528	0.4692	0.0986	0.0085	0.001	0.024
Reynolds-Stress JEPA	0.0111	0.0876 ± 0.0367	0.2700	0.4563	0.0842	**0.0023**	0.002	0.014
Potential-Flow JEPA	**0.0097**	0.0233 ± 0.0166	0.1775	0.3428	**0.0289**	0.0047	0.006	0.032
Observation-Space Reconstructive Baselines								
TimesNet (ICLR 2023)	0.2427	**0.2091 ± 0.0265**	**0.2931**	**0.5882**	0.2163	0.0563	**0.022**	**0.850**
TranAD (VLDB 2022)	0.2219	0.1405 ± 0.0154	0.2483	0.5334	0.2027	0.0398	0.018	0.811

TABLE IV
CALIBRATOR COMPARISON ON IDENTICAL SCORE STREAMS (33 STREAMS, BUFFERED INFERENCE, UP TO FIVE SEEDS). DESIGN RISK q=10^{-3}=0.1%; THE CI-EXCLUSION COLUMN SHOWS AUTOMATED CALIBRATION SYSTEMATICALLY OVERSHOOTS OUT-OF-SAMPLE.

Model Architecture	**Calibrator Rule**	**Held-Out Nominal FPR**	**CI Excludes $q = 10^{-3}$**	**Point-F1**
TS-JEPA (Control)	Standard SPOT ($q = 10^{-3}$)	**0.0526**	61%	0.1577
	SPOT + Kurtosis ($\beta = 0.167$)	0.1768	82%	**0.1694**
	Empirical 99.5th Percentile	0.0682	55%	0.1598
	Validation Maximum	0.0553	65%	0.1591
Operator-Entropy	Standard SPOT ($q = 10^{-3}$)	**0.0617**	65%	0.1599
	SPOT + Kurtosis ($\beta = 0.167$)	0.1229	76%	0.1441
	Empirical 99.5th Percentile	0.0758	62%	**0.1658**
	Validation Maximum	0.0656	64%	0.1627
TimesNet	Standard SPOT ($q = 10^{-3}$)	**0.2217**	86%	0.1395
	SPOT + Kurtosis ($\beta = 0.167$)	0.2281	82%	0.1352
	Empirical 99.5th Percentile	0.2338	87%	**0.1413**
	Validation Maximum	0.2263	86%	0.1410
TranAD	Standard SPOT ($q = 10^{-3}$)	**0.2217**	82%	0.0936
	SPOT + Kurtosis ($\beta = 0.167$)	0.2453	84%	0.0940
	Empirical 99.5th Percentile	0.2343	84%	**0.0962**
	Validation Maximum	0.2270	84%	0.0952

3) **Parity with empirical quantiles:** on TS-JEPA, validation-maximum gives FPR 0.0553 / F1 0.1591 and empirical p99.5 gives 0.0682 / 0.1598 — parametric GPD tail modeling confers no demonstrable benefit.
4) **Kurtosis-adaptive failure:** the multiplier inflates TS-JEPA FPR from 0.0526 to 0.1768 (CI-exclusion 61% $\rightarrow$ 82%) with indistinguishable F1 (0.1694 vs. 0.1577); calibration-split kurtosis does not forecast out-of-sample tail shifts.

To separate the causes of calibration failure, we recorded per-stream tail-fit diagnostics on a nine-stream non-GHL subset (TS-JEPA and TimesNet, three seeds each; `reports/evt_tail_diagnostics.csv`). Tail-sample scarcity is genuine — the median exceedance count above $u = p_{98}$ is only 21 (range 16–70), and the Grimshaw estimator returns the degenerate boundary shape $\xi = -0.9$ in 33% of TS-JEPA and 52% of TimesNet fits — yet goodness-of-fit is not the dominant failure: KS tests of the excesses reject at $p < 0.05$ in only $6/27$ (TS-JEPA) and $8/27$ (TimesNet) cells, while realized held-out FPR varies from 0.0 to 0.67 across seeds on the *same* stream under passing fits. Inter-block distribution drift, not tail mis-specification, drives the overshoot; exceedances are also strongly clustered (10–22 per cluster), inflating the effective variance of the FPR estimate. The four-mechanism decomposition and sensitivity sweep in Supplementary Section S-6 corroborate this attribution.

A controlled-drift study (Supplementary Table S19) injects mean shifts, variance rescaling, autocorrelation shifts, regime changes, and continuous ramps into the real dumped residual streams and re-evaluates five calibrators over multiple evaluation blocks. Two results sharpen the failure analysis: runs declustering leaves only 5–9 *effective* exceedances per cell (of a median 21 raw), so GPD tail fits rest on far fewer independent samples than the raw count suggests; and rolling calibrators control risk to 1–7% under step, variance, and autocorrelation shifts but fail under continuous ramps (0.33–0.39 achieved FPR): adaptation absorbs regime jumps but not drift within the evaluation block itself.

Computational profiling is in Supplementary Table S10: the JEPA family uses 183.8–185.3 MB peak memory (nearly 7$\times$ less than TranAD) at 0.60–0.84 ms per-window inference.

VIII. CONCLUSION AND FUTURE WORK

We investigated empirically whether JEPA-style latent prediction and physics-inspired regularizers resolve the false-alarm vulnerability of multivariate anomaly detection, evaluating 33 telemetry streams across seven physical domains under a strictly held-out calibration protocol. Three conclusions follow.

First, scoring in a contracted latent space conditions residual distributions: matched-backbone controls attribute the effect primarily to latent compression rather than to forecasting or the EMA target, filtering observation-space jitter to cut held-out nominal FPR roughly four-fold versus reconstructive baselines (-0.153 vs. TimesNet, p=0.002; -0.151 vs. TranAD, p=0.003, five-seed subset), while Point-F1 and PR-AUC remain statistically indistinguishable ($p \geq 0.13$). Under strictly causal streaming, latent prediction suppresses false alarms to 1.0%–1.9%; at matched false-alarm budgets its causal event recall instead exceeds the reconstructive baselines'.

Second, hypothesis testing refutes a detection advantage for the physics-inspired regularizers: neither Operator-Entropy nor Reynolds-Stress outperforms TS-JEPA ($p \geq 0.61$) across all 33 streams, and Potential-Flow is significantly worse (p=0.033), unable to represent circulating orbits. Matched-capacity controls confirm that disabling each physical term shifts Point-F1 by only $+0.001$ to $+0.006$ ($p \geq 0.58$); the formulations enforce their intended mathematics but real telemetry does not obey the idealized laws.

Third, SPOT calibration systematically overshoots its design risk on non-stationary telemetry (5.3–6.2% nominal FPR for JEPAs, $\approx$ 22% for baselines at q=10^{-3}; CIs exclude q in 61–86% of runs). Parametric tail modeling offers no advantage over empirical quantiles, and kurtosis adjustment degrades rather than corrects the drift.

Scope. All evaluations are *within-dataset*: models are trained, calibrated, and tested on disjoint chronological partitions of the same stream. We claim nothing about cross-dataset transfer or unseen fault types (cf. Section S-3.2). Extensions include solenoidal components via Helmholtz decomposition, memory-bounded online drift adaptation, accelerated streaming inference kernels, and integer quantization for edge deployment.

Code and Data Availability

All benchmark extraction scripts, preprocessing pipelines, checkpoints, training implementations, evaluation runners (`scripts/run_multiseed_heldout_benchmark.py`, `scripts/run_protocol_ablation.py`), diagnostic artifacts, and logs are released at https://anonymous.4open.science/r/NCAD_CS-Reproducibility-8F21; public datasets are accessible via their referenced repositories.

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